

QUESTION 11

In metamorphism of calc-silicate rocks, which assemblage can be considered equivalent to the sillimanite zone of pelitic rocks?

(a) Anorthite-pyroxene

(b) Zoisite-calcite-biotite

(c) Diopside-anorthite

✓ (d) Wollastonite-grossular

✓ CORRECT ANSWER: (d) Wollastonite-grossular

OPTION-BY-OPTION ANALYSIS

A ✗ INCORRECT

Anorthite-pyroxene

While anorthite does appear in calc-silicate rocks at higher grades, the "anorthite-pyroxene" assemblage is more characteristic of the

granulite facies

in mafic metamorphic rocks (amphibolite → granulite transition involves pyroxene replacing hornblende). It is not the recognised isograd-equivalent for the sillimanite zone in Kennedy's (1949) Scottish Highlands correlation. This assemblage is incomplete and does not correspond to any standard zone boundary in calc-silicates.

B ✗ INCORRECT

Zoisite-calcite-biotite

This is a

lower-grade assemblage

typical of the zoisite zone in calc-silicate rocks. In Ferry's (1983) Augusta-Waterville study, zoisite appears in the "zoisite zone" which corresponds to a grade *below* the diopside zone. The zoisite zone broadly correlates to the kyanite or lower-staurolite grade in pelites — well below the sillimanite zone. The presence of calcite also indicates lower metamorphic grade (calcite is consumed in high-grade reactions).

C ✗ INCORRECT

Diopside-anorthite

Diopside appears at the

diopside isograd

which, as Winter explicitly states, "occurs at about the grade of the sillimanite isograd in the pelites." While diopside is indeed a high-grade marker, the assemblage "diopside-anorthite" describes a *binary system* from igneous petrology (the classic Di-An eutectic system), not a metamorphic zone assemblage. Furthermore, in the Scottish Highlands correlation (Kennedy 1949), the sillimanite-equivalent in calc-silicates is specifically

wollastonite-grossular

, not diopside-anorthite. The diopside zone corresponds more broadly to the sillimanite zone onset, but "diopside-anorthite" is too generic and not the correct zone assemblage cited in the textbook.

D ✓ CORRECT

Wollastonite-grossular

This is the highest-grade calc-silicate assemblage that forms the **Garnet Zone**

in the Crestmore-type progression, with the reaction: vesuvianite + monticellite + diopside + wollastonite → grossular + diopside + wollastonite. Wollastonite forms from Cal + Qtz at temperatures >550°C at low P (and >750°C at higher P in closed systems), while grossular garnet (a Ca-Al garnet) is its highest-grade companion. Together, wollastonite + grossular characterise the peak metamorphic assemblage in calc-silicate rocks, directly equivalent to the sillimanite zone in pelites. Kennedy's (1949) Scottish Highlands study is the foundational reference establishing this correlation.

CONCEPT: METAMORPHISM OF CALC-SILICATE ROCKS — FULL CHAPTER OVERVIEW

Calc-silicate rocks are metamorphosed impure carbonates (limestones/dolostones with clay, quartz, feldspar). They develop a diagnostic sequence of mineral zones with increasing grade, just as pelitic rocks develop Barrovian zones. The correlation between pelitic and calc-silicate zones was first systematically established by **W. Q. Kennedy (1949)** in the Scottish Highlands, and later expanded by John Ferry (1983) in Maine.

CORE PRINCIPLE

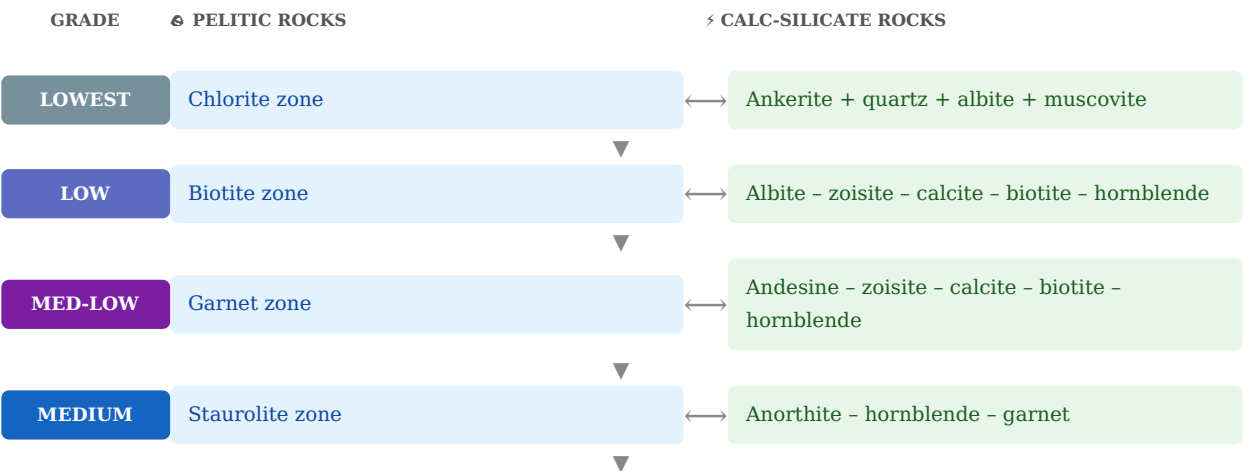
Because calc-silicate rocks and pelitic rocks co-exist in the same metamorphic terrane and experience the same P-T conditions, their mineral zones must be **isograde-equivalent** — they just reflect different bulk compositions responding to the same thermal gradient.

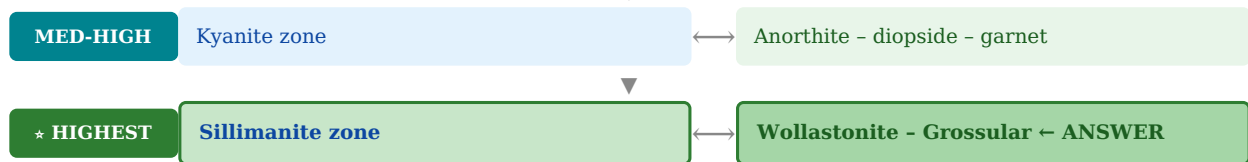
Kennedy's (1949) Pelitic ↔ Calc-Silicate Zone Correlation

After Kennedy (1949) and Tanner (1976), as cited in Winter (2010), Chapter 29, Table 1, p. 674

PELITIC ZONE (BARROVIAN)	CALC-SILICATE MINERAL ASSEMBLAGE	GRADE
Chlorite	—	Lowest
Biotite	Albite - zoisite - calcite - biotite - hornblende	Low
Garnet	Andesine - zoisite - calcite - biotite - hornblende	Lower-medium
Staurolite	Anorthite - hornblende - garnet	Medium
Kyanite	Anorthite - diopside - garnet	Medium-high
Sillimanite	Wollastonite - grossular	Highest

PROGRESSIVE METAMORPHISM: GRADE LADDER (CALC-SILICATE VS. PELITIC)

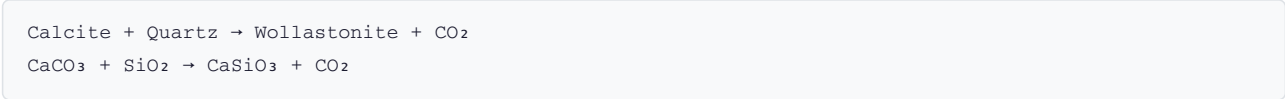




HOW WOLLASTONITE AND GROSSULAR FORM (KEY REACTIONS)

Wollastonite Formation

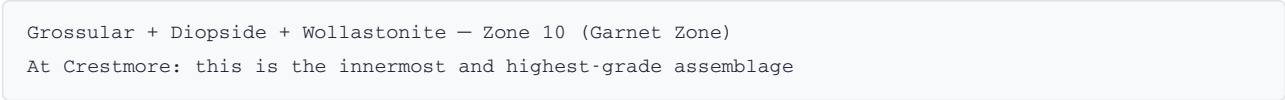
Wollastonite (CaSiO_3) is produced by the decarbonation of calcite with quartz — the "wollastonite isograd reaction":



This reaction occurs at $T > 550^\circ\text{C}$ at $P = 0.1 \text{ GPa}$ (if $X_{\text{CO}_2} > 0.2$), but may require $>750^\circ\text{C}$ in closed systems buffered to high X_{CO_2} . The wollastonite isograd is a classic **decarbonation reaction**.

Grossular Formation

Grossular garnet ($\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$) forms at high-grade conditions in calc-silicate rocks, as the highest-grade zone marker:



*"In this progression of zones, we can again see the sequential development of index minerals toward the igneous contact, such as clinohumite, followed by forsterite, clintonite, monticellite, melilite, spurrite/tilleyite, merwinite, vesuvianite, diopside, **wollastonite, and finally grossular garnet.**"*

— Winter (2010), Principles of Igneous and Metamorphic Petrology, Chapter 29 (Crestmore Aureole discussion, p. 486)

*"The diopside isograd in the calcareous rocks occurs at about the grade of the **sillimanite isograd in the pelites**. Diopside growth results from dehydration of amphibole at high grade..."*

— Winter (2010), Chapter 29, §1.4.5 Diopside Zone, p. 674

"Wollastonite may form in the upper amphibolite facies or in the granulite facies. Because of the high temperatures required, it is not common in regional metamorphic rocks."

— Winter (2010), Chapter 29, §1.2 Regional Metamorphism, p. 670

CONTACT VS. REGIONAL METAMORPHISM OF CALC-SILICATES

ASPECT	CONTACT METAMORPHISM	REGIONAL METAMORPHISM
Heat source	Igneous intrusion (pluton)	Geothermal/orogenic burial
Typical example	Crestmore, California; Alta, Utah	Scottish Highlands, Maine (Waterville)
Zone width	Narrow aureole (tens of metres to km)	Broad regional zones (tens of km)
Low-grade minerals	Talc, tremolite (in dolostones)	Tremolite (talc less common)
Highest-grade minerals	Wollastonite, grossular, periclase, monticellite	Wollastonite, diopside, grossular
Fluid system	Often open (H_2O infiltration from pluton)	Often closed (internally buffered)

P-T diagram used	T-X _{CO₂} at low P (0.1 GPa)	T-X _{CO₂} at higher P (0.5 GPa)
Sillimanite-equivalent zone	Wollastonite - Grossular (BOTH cases)	

CLASSIC CALC-SILICATE ZONE SEQUENCE — CRESTMORE, CALIFORNIA

Burnham (1959) mapped the following zones in the carbonate aureole at Crestmore, CA. This is the classic reference sequence cited by Winter:

ZONE NO.	ZONE NAME	ASSEMBLAGE
1-3	Forsterite Zone	Calcite + brucite/clinochumite/forsterite + spinel
4-7	Monticellite Zone	Calcite + forsterite/monticellite + spurrite/merwinite + melilite
8	Vesuvianite Zone	Vesuvianite + monticellite + spurrite + merwinite + melilite
9	(Upper Vesuvianite)	Vesuvianite + monticellite + diopside + wollastonite
10	Garnet Zone	Grossular + diopside + wollastonite ← HIGHEST GRADE

🔑 EXAM KEY POINT

Zone 10 (Grossular + Diopside + Wollastonite) is the innermost, highest-grade zone at Crestmore. Wollastonite and grossular are the diagnostic minerals of the peak calc-silicate assemblage, equivalent to sillimanite in pelites.

QUICK MINERAL REFERENCE FOR CALC-SILICATE ROCKS

MINERAL	FORMULA	GRADE INDICATOR	IN PELITIC EQUIVALENT?
Calcite	CaCO ₃	All grades; consumed at high grade	Not a pelitic mineral
Talc	Mg ₃ Si ₄ O ₁₀ (OH) ₂	Very low grade (in dolostones)	Chlorite zone analogue
Tremolite	Ca ₂ Mg ₅ Si ₈ O ₂₂ (OH) ₂	Low-medium grade	Biotite zone analogue
Zoisite / Epidote	Ca ₂ Al ₃ Si ₃ O ₁₂ (OH)	Lower-medium grade (H ₂ O-rich)	Garnet zone analogue
Diopside	CaMgSi ₂ O ₆	Medium-high grade (key isograd)	Kyanite-sillimanite zone
Wollastonite	CaSiO₃	High grade (T > 550°C); peak marker	Sillimanite zone
Grossular	Ca₃Al₂Si₃O₁₂	High grade; innermost zone	Sillimanite zone
Vesuvianite	Complex Ca-Mg-Al silicate	High-T contact aureoles (sanidinite)	No exact pelitic equivalent
Monticellite	CaMgSiO ₄	Very high T (> 800°C); rare	Granulite facies analogue

THE CAO-MGO-SIO₂ (CMS) SYSTEM — CHEMICAL FRAMEWORK

Most calc-silicate mineral assemblages are understood through the **CaO-MgO-SiO₂-H₂O-CO₂ (CMS-HC)** system. When projected from CO₂ and H₂O, the minerals plot in a triangular CMS diagram, and the progressive zones correspond to increasing SiO₂ activity (from intrusion-derived fluids or original impurities).

CaO corner → Calcite, Dolomite

MgO corner → Periclase, Brucite

SiO₂ corner → Quartz

Wollastonite → near CaO-SiO₂ join

Diopside → CaO-MgO-SiO₂ midfield

"Wollastonite occurs only in limestones and very siliceous dolostones and is separated from virtually all of the dark shaded common dolostone area by the Cal-Di and Cal-Tr tie-lines. In most dolostones, quartz is consumed by early reaction with dolomite to create these tie-lines."

— Winter (2010), Chapter 29, p. 667

The **T-X_{CO₂} diagram** is the primary tool used for understanding calc-silicate metamorphism. Because both CO₂ and H₂O are released in decarbonation and dehydration reactions, the fluid composition (X_{CO₂}) is a critical variable, unlike in pelitic systems where only H₂O is dominant.

PROBABLE EXAM QUESTIONS FROM THIS TOPIC

Q1. The wollastonite isograd in calc-silicate rocks is defined by which reaction?

→ **Answer: Calcite + Quartz → Wollastonite + CO₂. This is a decarbonation reaction requiring T > 550°C at low P.**

Q2. In which metamorphic facies would you expect to find both wollastonite and grossular garnet in a calc-silicate rock?

→ **Answer: Upper amphibolite facies to granulite facies (equivalent to sillimanite zone in pelites).**

Q3. Which zone in calc-silicate rocks is equivalent to the kyanite zone in pelitic rocks?

→ **Answer: Anorthite-diopside-garnet (Kennedy 1949; Scottish Highlands correlation).**

Q4. What are the five progressive zones recognised by Ferry (1983) in calcareous rocks of the Augusta-Waterville area, Maine?

→ **Answer: (1) Ankerite zone, (2) Biotite zone, (3) Amphibole zone, (4) Zoisite zone, (5) Diopside zone. The diopside zone corresponds to the sillimanite isograd in pelites.**

Q5. In a siliceous dolostone contact aureole, which mineral is produced by the reaction: 11 Dolomite + Tremolite → 8 Forsterite + 13 Calcite + H₂O + 9 CO₂?

→ **Answer: Forsterite. This is the forsterite-in isograd reaction in dolostones (Reaction 9, Winter Chapter 29).**

Q6. A T-X_{CO₂} diagram is preferred over a T-P diagram for studying metamorphism of which rock type, and why?

→ **Answer: Calcareous/calc-silicate rocks. Because fluid composition (X_{CO₂}) is a critical variable — both CO₂ and H₂O are generated by reactions, making fluid composition as important as temperature.**

Q7. Which study by Kennedy (1949) is classically cited for establishing the equivalence of calc-silicate mineral zones with Barrovian pelitic zones?

→ **Answer: Kennedy's study of metamorphosed calc-silicate rocks in the Scottish Highlands, summarised in Table 1 of Winter (2010), Chapter 29.**

Q8. In a closed metamorphic system, why does the wollastonite isograd occur at higher temperatures than in an open system?

→ **Answer: Because earlier reactions buffer the fluid to high X_{CO₂} values (>0.62), and since Cal + Qtz → Wo + CO₂ requires low X_{CO₂} to proceed, the high CO₂ activity shifts the reaction to much higher temperatures (>750°C vs ~550°C in open systems).**

Q9. The assemblage at Crestmore Zone 10 (innermost) is grossular + diopside + wollastonite. What does this represent petrogenetically?

→ **Answer: The highest-grade, innermost calc-silicate zone in the contact aureole, equivalent to the sillimanite zone in pelites. Grossular garnet is the last index mineral to appear in the progressive sequence.**

Q10. Which calc-silicate assemblage is characteristic of the biotite-grade equivalent zone?

→ **Answer: Albite - zoisite - calcite - biotite - hornblende (Kennedy 1949).**

SUMMARY: WHY EACH OPTION IS RIGHT/WRONG

OPTION	ASSEMBLAGE	GRADE IN CALC-SILICATES	PELITIC EQUIVALENT	VERDICT
(a)	Anorthite-pyroxene	Granulite facies (mafic rocks)	Not a standard Barrovian zone marker	✗ Wrong
(b)	Zoisite-calcite-biotite	Low-medium grade (Zoisite Zone)	Garnet / lower staurolite zone	✗ Wrong
(c)	Diopside-anorthite	A binary igneous system, not a defined calc-silicate zone	Kyanite-sillimanite boundary (partial)	✗ Wrong
(d)	Wollastonite-grossular	Highest grade — Garnet/Wollastonite Zone	Sillimanite zone ✓	✓ Correct

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Source: Winter, J.D. (2010). *Principles of Igneous and Metamorphic Petrology*, 2nd Ed. Chapter 29 — Metamorphism of Calcareous and Ultramafic Rocks. Table 1 (Kennedy 1949) · §1.4.5 · p. 674, 670, 486

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